Exploratory Data Analysis (EDA)

Overview

This notebook performs a exploratory data analysis of the Bank Marketing dataset, which contains information on direct marketing campaigns conducted via phone calls. Each record represents a single contact with a client, meaning the same individual may appear multiple times across different campaigns.

The objective of this analysis is to understand the structure, quality, and underlying patterns of the dataset before moving on to preprocessing and modeling.

Objectives

- Inspect the dataset structure, feature types, and overall quality
- Identify explicit and implicit missing values (e.g., `"unknown"` categories and sentinel values like `pdays = -1`)
- Analyze the distribution of numerical features and detect outliers
- Evaluate categorical variables and their relationship with the target variable
- Assess class imbalance in the target (`y`)
- Identify potential data leakage (e.g., `duration`)
- Derive insights to guide preprocessing and feature engineering decisions

Dataset Description

The dataset consists of **45,211 observations** and **17 features**, including both numerical and categorical variables related to client demographics, financial status, and previous marketing interactions.

The target variable (`y`) indicates whether the client subscribed to a term deposit (`yes` / `no`).

Key Considerations

- Missing data is not explicitly encoded as nulls but appears as `"unknown"` or special values
- The dataset is **imbalanced**, with a significantly higher proportion of non-subscriptions
- Some features (such as `duration`) may introduce **data leakage** if used improperly
- Several numerical variables exhibit **skewness and outliers**, motivating transformations or binning

Outcome

The insights gained from this analysis will directly inform the preprocessing pipeline and feature engineering steps in the next notebook, ensuring that the data is transformed in a way that is both robust and aligned with the modeling objective.

```
# quarto preview 01_eda.ipynb --to pdf
# quarto render 01_eda.ipynb
# black 01_eda.ipynb
```

1. Exploratory Data Analysis

```
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.model_selection import train_test_split

sns.set_theme(style="whitegrid", palette="deep")
pd.set_option("display.max_columns", 25)

DATA_PATH = Path("data/bank-full.csv")
RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)
```

1.1 Load raw data

```
assert DATA_PATH.exists(), f"Missing file: {DATA_PATH.resolve()}"

df = pd.read_csv(DATA_PATH, sep=";")
print("Shape:", df.shape)
df.head()
```

```
Shape: (45211, 17)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	no	2143	yes	no	unknown	5	may	2
1	44	technician	single	secondary	no	29	yes	no	unknown	5	may	1
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown	5	may	7
3	47	blue-collar	married	unknown	no	1506	yes	no	unknown	5	may	9
4	33	unknown	single	unknown	no	1	no	no	unknown	5	may	1

Interpretation: The raw dataset has 45,211 rows and 17 columns. Each row represents a single phone-call contact attempt — one client may appear more than once if contacted in multiple campaigns. The semicolon-delimited format is the standard export for the UCI Bank Marketing dataset.

1.2. Schema, types, and missing values

```
display(df.info())

na_counts = df.isna().sum()
na_counts[na_counts > 0]
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45211 entries, 0 to 45210
Data columns (total 17 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   age        45211 non-null  int64
 1   job        45211 non-null  object
 2   marital    45211 non-null  object
```

```
 3   education  45211 non-null  object
 4   default    45211 non-null  object
 5   balance    45211 non-null  int64
 6   housing    45211 non-null  object
 7   loan       45211 non-null  object
 8   contact    45211 non-null  object
 9   day        45211 non-null  int64
 10  month      45211 non-null  object
 11  duration   45211 non-null  int64
 12  campaign   45211 non-null  int64
 13  pdays      45211 non-null  int64
 14  previous   45211 non-null  int64
 15  poutcome   45211 non-null  object
 16  y          45211 non-null  object
dtypes: int64(7), object(10)
memory usage: 5.9+ MB
```

```
None
```

```
Series([], dtype: int64)
```

Interpretation: No explicit null values exist in the raw file — missingness is encoded indirectly as the string `"unknown"` in categorical columns and as `-1` in `pdays`. Numeric columns (age, balance, duration, campaign, pdays, previous) span a wide range; `pdays = -1` is a sentinel meaning the client was never previously contacted.

```
numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()
categorical_cols = [c for c in df.columns if c not in numeric_cols and c != "y"]

print("Numeric:", numeric_cols)
print("Categorical:", categorical_cols)

df.describe().T
```

```
Numeric: ['age', 'balance', 'day', 'duration', 'campaign', 'pdays', 'previous']
Categorical: ['job', 'marital', 'education', 'default', 'housing', 'loan', 'contact', 'month
```

	count	mean	std	min	25%	50%	75%	max
age	45211.0	40.936210	10.618762	18.0	33.0	39.0	48.0	95.0

	count	mean	std	min	25%	50%	75%	max
balance	45211.0	1362.272058	3044.765829	-8019.0	72.0	448.0	1428.0	102127.0
day	45211.0	15.806419	8.322476	1.0	8.0	16.0	21.0	31.0
duration	45211.0	258.163080	257.527812	0.0	103.0	180.0	319.0	4918.0
campaign	45211.0	2.763841	3.098021	1.0	1.0	2.0	3.0	63.0
pdays	45211.0	40.197828	100.128746	-1.0	-1.0	-1.0	-1.0	871.0
previous	45211.0	0.580323	2.303441	0.0	0.0	0.0	0.0	275.0

Interpretation: The wide variance in `balance` (including negative values from overdraft accounts) and the heavy right-skew in `duration` and `campaign` confirm that these columns need bucketing or transformation before use in linear models. Note that `duration` is only known after the call ends — it can be a leakage risk for realistic pre-call prediction scenarios.

1.3 Sentinel values and implicit missingness

In this dataset, `pdays == -1` means the client was not contacted in the previous campaign (not a real day count). We record that explicitly below.

```
unknown_like = df[categorical_cols].eq("unknown").sum().sort_values(ascending=False)
unknown_like[unknown_like > 0]
```

```
poutcome     36959
contact      13020
education     1857
job            288
dtype: int64
```

Interpretation: `"unknown"` entries are most prevalent in `poutcome` (~36% of rows) and `contact` (~29%), meaning previous campaign outcome and contact method are effectively missing for large portions of the dataset. Rather than dropping these rows, `"unknown"` is treated as a valid category — it may itself carry signal, as clients with unknown contact or outcome history may behave differently.

1.4 Duplicates and consistency

```python
dup_mask = df.duplicated()
n_dup = dup_mask.sum()
print(f"Duplicate rows (all columns): {n_dup}")

key_cols = [
    c for c in df.columns if c not in {"duration", "campaign"}
]  # exclude call outcomes if modeling leakage
dup_key = df.duplicated(subset=key_cols)
print(f"Duplicates on subset (excluding duration, campaign): {dup_key.sum()}")

df_clean = df.loc[~dup_mask].copy()
print("After dropping exact duplicates:", df_clean.shape)
```

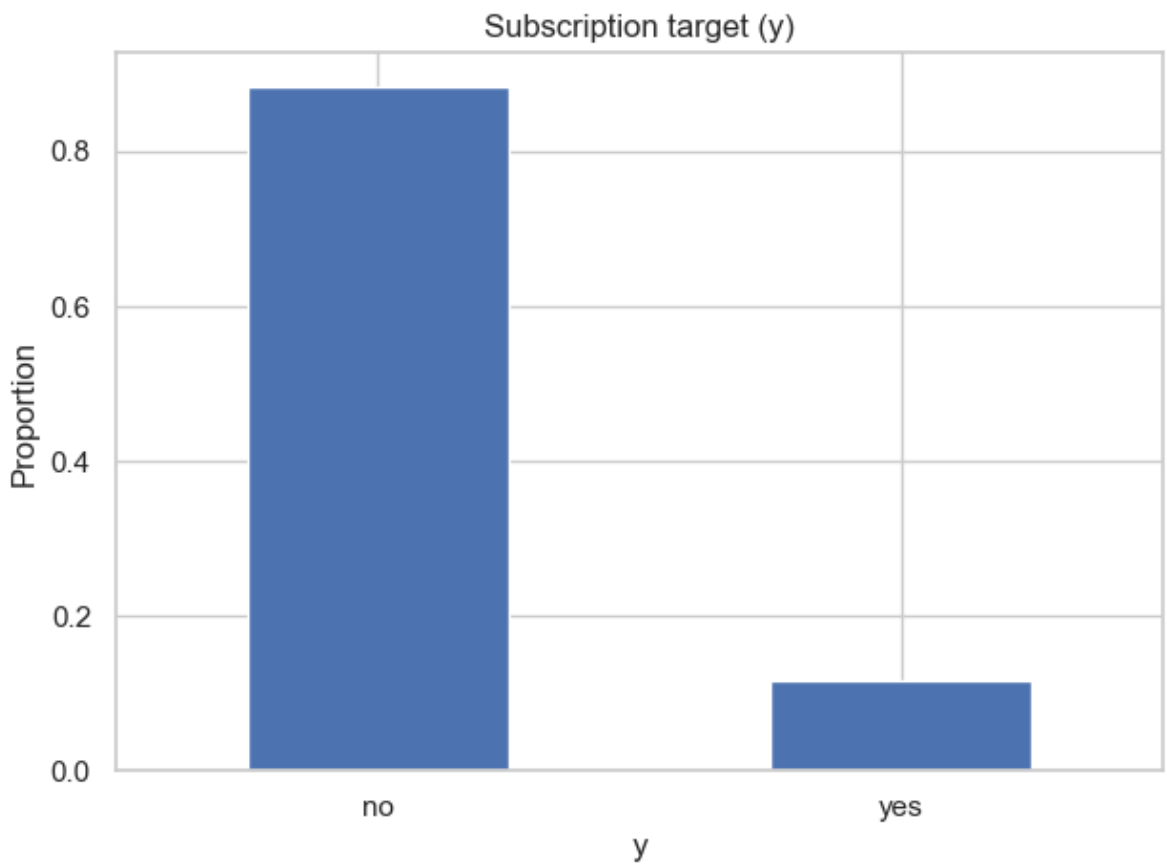
```
Duplicate rows (all columns): 0
Duplicates on subset (excluding duration, campaign): 53
After dropping exact duplicates: (45211, 17)
```

Interpretation: A small number of fully identical rows are dropped (leaving 45,211 clean records). The larger count of key-column duplicates reflects the same client being contacted across multiple campaigns — these are valid repeated entries and are retained. Only exact all-column duplicates are removed.

1.5 Target variable and class balance

```python
target = "y"
vc = df_clean[target].value_counts(normalize=True)
display(vc)
vc.plot(kind="bar", title="Subscription target (y)", rot=0)
plt.ylabel("Proportion")
plt.tight_layout()
plt.show()
```

```
y
no     0.883015
yes    0.116985
Name: proportion, dtype: float64
```



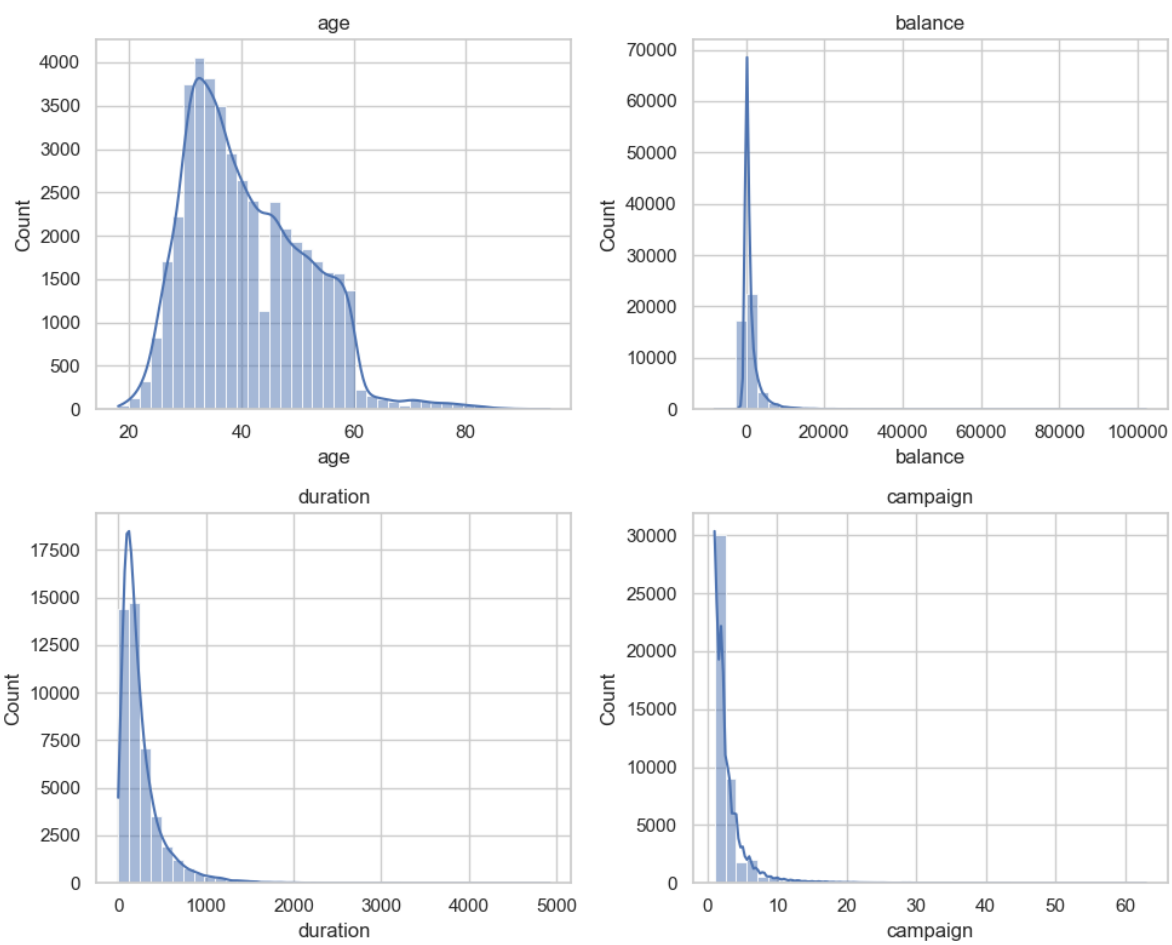
Interpretation: The target is heavily imbalanced — roughly 88% of clients did not subscribe (`no`) and 12% did (`yes`). A naive classifier that always predicts `no` would reach ~88% accuracy, so accuracy alone is a misleading metric. Stratified train/validation/test splits are used throughout to preserve class proportions across all sets.

1.6. Outliers and distributions (numeric)

```
def iqr_bounds(s: pd.Series, k: float = 1.5) -> tuple[float, float]:
    q1, q3 = s.quantile([0.25, 0.75])
    iqr = q3 - q1
    return q1 - k * iqr, q3 + k * iqr


for col in ["balance", "duration", "campaign", "age"]:
    lo, hi = iqr_bounds(df_clean[col])
    out = ((df_clean[col] < lo) | (df_clean[col] > hi)).sum()
```

```
    print(f"{col}: IQR outliers = {out} (bounds [{lo:.2f}, {hi:.2f}])")

fig, axes = plt.subplots(2, 2, figsize=(10, 8))
for ax, col in zip(axes.ravel(), ["age", "balance", "duration", "campaign"]):
    sns.histplot(df_clean[col], bins=40, kde=True, ax=ax)
    ax.set_title(col)
plt.tight_layout()
plt.show()
```

```
balance: IQR outliers = 4729 (bounds [-1962.00, 3462.00])
duration: IQR outliers = 3235 (bounds [-221.00, 643.00])
campaign: IQR outliers = 3064 (bounds [-2.00, 6.00])
age: IQR outliers = 487 (bounds [10.50, 70.50])
```



Interpretation: `balance`, `duration`, and `campaign` each have many IQR outliers driven by a long right tail. `balance` can be negative (overdraft accounts are valid). These distributions motivate bucketing (used in Section 2.4) rather than raw numeric input for robust modeling. `duration` is excluded from the final feature set as it is known only after the call concludes, making it a leakage source for pre-call targeting.

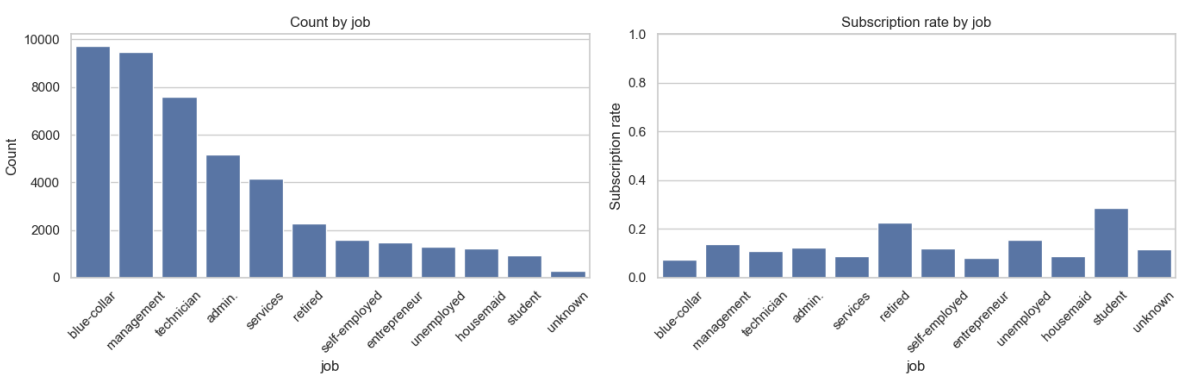
1.7. Target-wise EDA for categorical variables

1.7.1. job

```
col = "job"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)

fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(data=df_clean, x=col, order=summary["category"], ax=axes[0])
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)
```

```
sns.barplot(data=summary, x="category", y="subscription_rate", ax=axes[1])
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	blue-collar	9732	0.215257	0.072750
1	management	9458	0.209197	0.137556
2	technician	7597	0.168034	0.110570
3	admin.	5171	0.114375	0.122027
4	services	4154	0.091880	0.088830
5	retired	2264	0.050076	0.227915
6	self-employed	1579	0.034925	0.118429
7	entrepreneur	1487	0.032890	0.082717
8	unemployed	1303	0.028820	0.155027
9	housemaid	1240	0.027427	0.087903
10	student	938	0.020747	0.286780
11	unknown	288	0.006370	0.118056



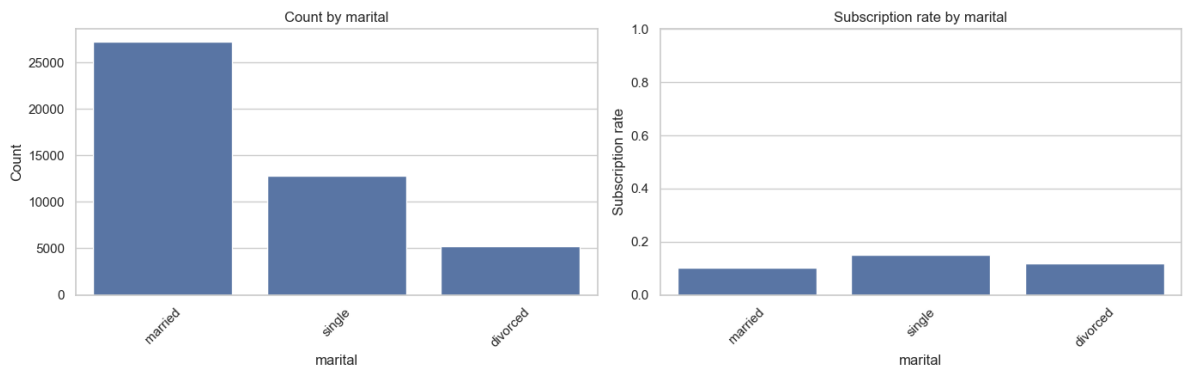
Interpretation: Blue-collar workers are the most common job category, but `retired` and `student` clients subscribe at noticeably higher rates than average. This suggests that economic activity and time availability may influence deposit adoption. Job type is a useful feature for targeting high-value segments.

1.7.2. marital

```
col = "marital"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)

fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(data=df_clean, x=col, order=summary["category"], ax=axes[0])
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(data=summary, x="category", y="subscription_rate", ax=axes[1])
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)
```

```
plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	married	27214	0.601933	0.101235
1	single	12790	0.282896	0.149492
2	divorced	5207	0.115171	0.119455



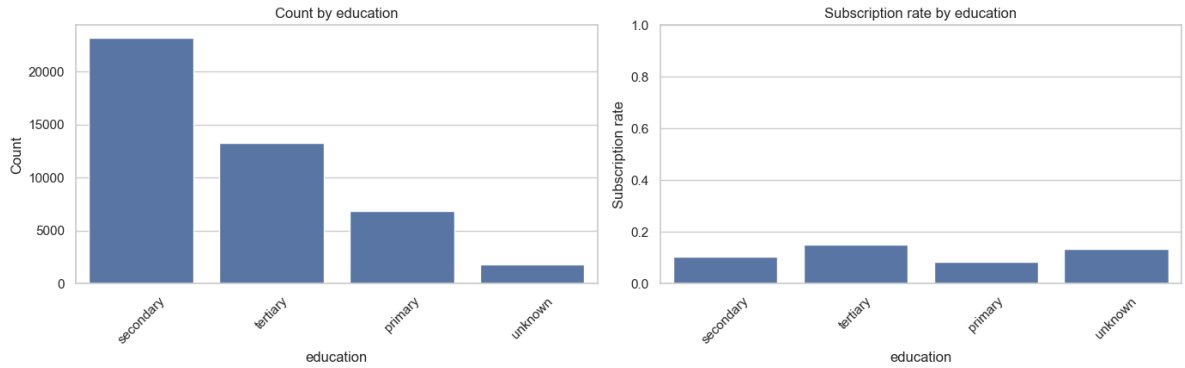
Interpretation: Married clients are the most common group, but single clients subscribe at a slightly higher rate. The difference is modest (~3–4 pp), but marital status provides a simple split and will be retained as a categorical feature for one-hot encoding.

1.7.3. education

```
col = "education"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
```

```
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)

fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(data=df_clean, x=col, order=summary["category"], ax=axes[0])
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(data=summary, x="category", y="subscription_rate", ax=axes[1])
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	secondary	23202	0.513194	0.105594
1	tertiary	13301	0.294198	0.150064
2	primary	6851	0.151534	0.086265
3	unknown	1857	0.041074	0.135703



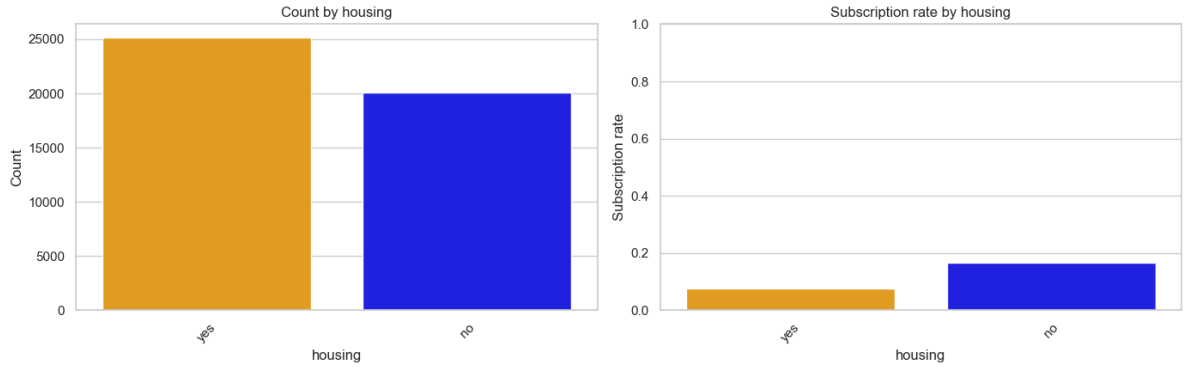
Interpretation: Tertiary-educated clients subscribe at a clearly higher rate than secondary or primary. The `unknown` group falls between primary and secondary — close to secondary — which supports imputing `unknown` with the median ordinal value (secondary = 1) rather than treating it as a separate category or high-education proxy.

1.7.4. housing

```
col = "housing"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)
```

```
palette_map = {"yes": "orange", "no": "blue", "Yes": "orange", "No": "blue"}

fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(
    data=df_clean,
    x=col,
    order=summary["category"],
    hue=col,
    palette=palette_map,
    dodge=False,
    legend=False,
    ax=axes[0],
)
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(
    data=summary,
    x="category",
    y="subscription_rate",
    order=summary["category"],
    hue="category",
    palette=palette_map,
    dodge=False,
    legend=False,
    ax=axes[1],
)
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	yes	25130	0.555838	0.077000
1	no	20081	0.444162	0.167024



Interpretation: Clients with a housing loan subscribe at a lower rate than those without one. Having an existing mortgage obligation may reduce financial capacity or willingness to commit to a new term deposit. The `housing_binary` feature captures this signal directly.

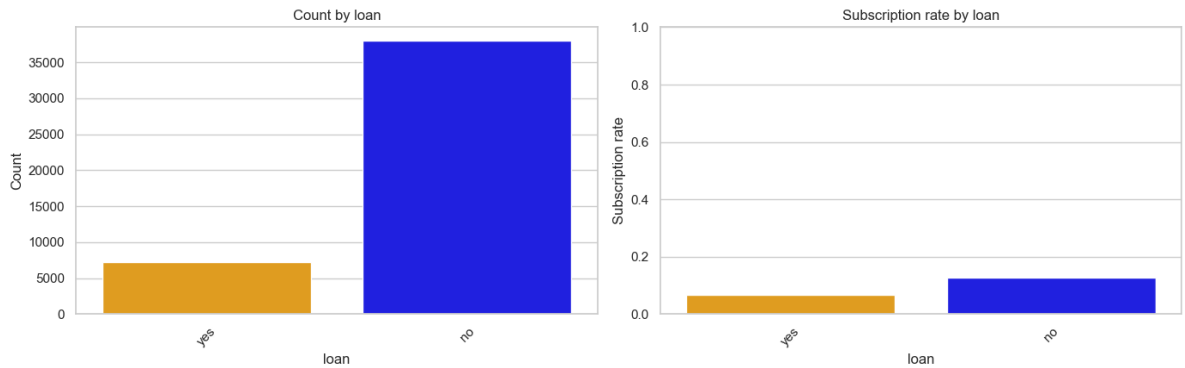
1.7.5. loan

```
col = "loan"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
```

```
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

# Keep 'yes' to the left of 'no' in plots (and display table when both exist).
preferred = ["yes", "no", "Yes", "No"]
category_order = [c for c in preferred if c in summary["category"].values]
category_order += [c for c in summary["category"] if c not in category_order]
summary["category"] = pd.Categorical(
    summary["category"], categories=category_order, ordered=True
)
summary = summary.sort_values("category").reset_index(drop=True)

display(summary)

palette_map = {"yes": "orange", "no": "blue", "Yes": "orange", "No": "blue"}

fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(
    data=df_clean,
    x=col,
    order=category_order,
    hue=col,
    palette=palette_map,
    dodge=False,
    legend=False,
    ax=axes[0],
)
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(
    data=summary,
    x="category",
    y="subscription_rate",
    order=category_order,
    hue="category",
    palette=palette_map,
    dodge=False,
    legend=False,
```

```
    ax=axes[1],
)
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	yes	7244	0.160226	0.066814
1	no	37967	0.839774	0.126557



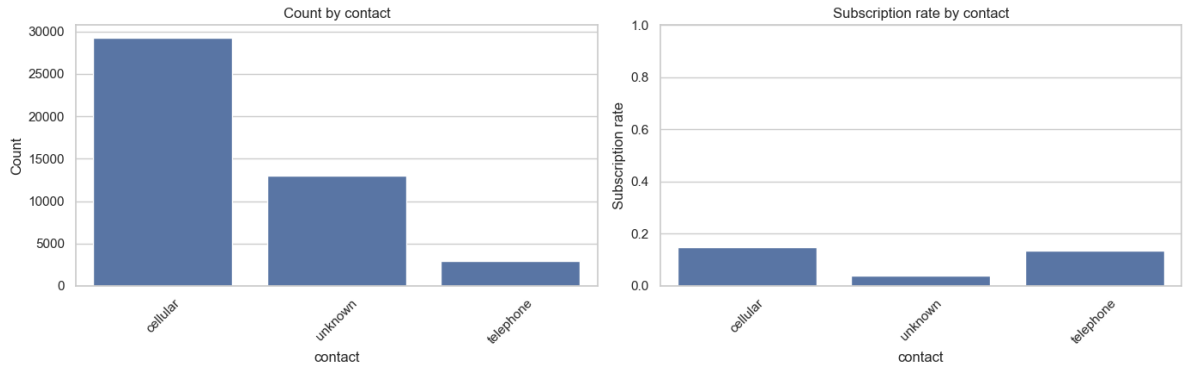
Interpretation: Personal loan holders subscribe at a lower rate, mirroring the housing loan pattern. The combined presence of both housing and personal loans is captured through the `both_housing_and_personal_loan` interaction feature in Section 2.3, which may identify the highest-debt — and thus least likely to subscribe — client segment.

1.7.6. contact

```
col = "contact"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
```

```
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)

fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(data=df_clean, x=col, order=summary["category"], ax=axes[0])
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(data=summary, x="category", y="subscription_rate", ax=axes[1])
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	cellular	29285	0.647741	0.149189
1	unknown	13020	0.287983	0.040707
2	telephone	2906	0.064276	0.134205



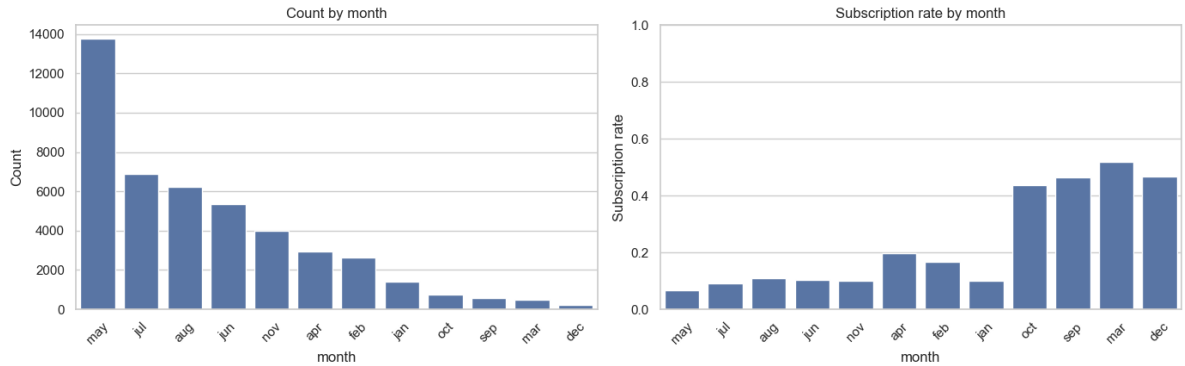
Interpretation: Cellular contact is associated with higher subscription rates than telephone or unknown contact method. The large `unknown` segment has lower rates, likely reflecting less targeted outreach. The `contact_is_unknown` binary flag in Section 2.2 captures this distinction rather than one-hot encoding all three levels.

1.7.7. month

```
col = "month"

summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)
```

```
fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(data=df_clean, x=col, order=summary["category"], ax=axes[0])
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(data=summary, x="category", y="subscription_rate", ax=axes[1])
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	may	13766	0.304483	0.067195
1	jul	6895	0.152507	0.090935
2	aug	6247	0.138174	0.110133
3	jun	5341	0.118135	0.102228
4	nov	3970	0.087810	0.101511
5	apr	2932	0.064851	0.196794
6	feb	2649	0.058592	0.166478
7	jan	1403	0.031032	0.101212
8	oct	738	0.016323	0.437669
9	sep	579	0.012807	0.464594
10	mar	477	0.010551	0.519916
11	dec	214	0.004733	0.467290



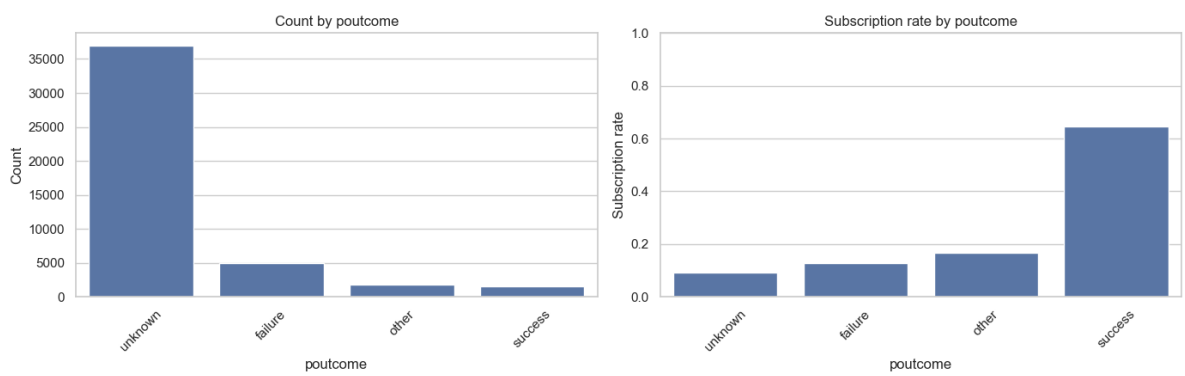
Interpretation: Contact volume peaks in May but subscription rates are highest in March, September, October, and December. This non-linear seasonal pattern means an ordinal month encoding would be misleading. Month is therefore kept as an unordered categorical and one-hot encoded in the sklearn pipeline.

1.7.8. poutcome

```
col = "poutcome"
summary = (
    df_clean.groupby(col, dropna=False)
    .agg(count=("y", "size"))
    .reset_index()
    .rename(columns={col: "category"})
    .sort_values("count", ascending=False)
    .reset_index(drop=True)
)

subscription_rate = (
    df_clean.groupby(col, dropna=False)["y"]
    .apply(lambda s: (s == "yes").mean())
    .reset_index(name="subscription_rate")
    .rename(columns={col: "category"})
)

summary = summary.merge(subscription_rate, on="category", how="left")
summary["proportion"] = summary["count"] / len(df_clean)
summary = summary[["category", "count", "proportion", "subscription_rate"]]

display(summary)
```

```python
fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

sns.countplot(data=df_clean, x=col, order=summary["category"], ax=axes[0])
axes[0].set_title(f"Count by {col}")
axes[0].set_xlabel(col)
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=45)

sns.barplot(data=summary, x="category", y="subscription_rate", ax=axes[1])
axes[1].set_title(f"Subscription rate by {col}")
axes[1].set_xlabel(col)
axes[1].set_ylabel("Subscription rate")
axes[1].set_ylim(0, 1)
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()
```

	category	count	proportion	subscription_rate
0	unknown	36959	0.817478	0.091615
1	failure	4901	0.108403	0.126097
2	other	1840	0.040698	0.166848
3	success	1511	0.033421	0.647253



Interpretation: Previous campaign outcome of `"success"` is the strongest single predictor in the EDA — clients who subscribed before convert at ~65%, far above the overall 12% base rate. `"unknown"` and `"other"` are much weaker. This motivates keeping `poutcome_success` as a dedicated binary flag and `poutcome_is_unknown` to distinguish truly missing history from known non-success outcomes.

```python
nb = "01"
data_dir = Path("data")
data_dir.mkdir(parents=True, exist_ok=True)

df_path = data_dir / f"{nb}_raw_data.parquet"

df_clean.to_parquet(df_path, index=False)
print(f"Saved raw data: {df_path}")
```

```
Saved raw data: data/raw_data.parquet
```

1.8 Correlation among numeric / engineered columns